

Chapter 1

The Family of Vehicle Routing Problems

Vehicle Routing: Problems, Methods, and Applications
2nd Edition, SIAM 2014

Stefan Irnich Paolo Toth Daniele Vigo

Lecture Slides — Self-Contained Reader Edition

SIAM, 2014

Chapter Overview

- 1 1.1 Introduction
- 2 1.2 The Capacitated Vehicle Routing Problem (CVRP)
- 3 1.3 The Family of VRP
- 4 1.4 Relationship to TSP and Bin Packing
- 5 1.5 Benchmark Instances
- 6 1.6 Objectives and Performance Metrics
- 7 Algorithms: Clarke-Wright and 2-Opt
- 8 Summary
- 9 Bibliography Highlights

What Is a Vehicle Routing Problem? I

The **family of vehicle routing problems (VRP)** encompasses one of the most practically important and theoretically rich classes of combinatorial optimisation. A single verbal definition unifies all variants:

Generic Definition (Irnich, Toth, Vigo 2014)

Given: A set of *transportation requests* and a *fleet of vehicles*.

Task: Determine a set of *vehicle routes* to **perform** all (or some) transportation requests with the given vehicle fleet **at minimum cost**; in particular, decide which vehicle handles which requests in which sequence so that all vehicle routes can be *feasibly* executed.

In problems subsumed under the VRP umbrella, transportation requests are **concentrated at specific points** of a road network (e.g. customer addresses, warehouses, hospitals). This distinguishes the VRP from the *Arc Routing Problem (ARP)*, in which requests are dispersed along road *segments* (e.g. street sweeping, postal delivery along streets).

What Is a Vehicle Routing Problem? II

The four fundamental VRP components are:

- 1 **Transportation requests** — what needs to be moved, how much, between which points, and under what time or sequence constraints.
- 2 **Fleet** — how many vehicles, what type, where they start and end, and what their capacities and operating costs are.
- 3 **Costs and profits** — the objective to be optimised, which may include travel distance, time, fuel, driver wages, or collected revenue.
- 4 **Feasibility of routes** — the constraints that a plan must satisfy, such as capacity limits, time windows, precedence requirements, and legal driving-hour regulations.

Key Insight

The enormous practical relevance of VRP stems from the fact that computerised optimisation methods yield **substantial savings in global transportation costs**. Even a 5% improvement over manual planning on a large logistics network translates to millions of euros per year.

What Is a Vehicle Routing Problem? III

Economic and scientific motivation. The success of VRP solution software rests on two pillars. First, modern computers can execute billions of arithmetic operations per second, enabling the evaluation of thousands of candidate routes in practical run times. Second, and equally important, the development of rigorous *mathematical models* has made it possible to describe all relevant real-world constraints exactly, so that the optimiser cannot “cheat” by exploiting model gaps.

This chapter introduces the building blocks that appear in virtually every VRP variant: the graph model of the road network, the mathematical formulation of capacity constraints, and the taxonomy of how time, stochasticity, fleet heterogeneity, and other features specialise the basic problem into distinct sub-problems. Every subsequent chapter of the book refines one or more of these building blocks.

Historical Note

The term *Vehicle Routing Problem* was coined by Dantzig and Ramser in 1959, who solved the fuel-delivery problem for a gas company. Since then, over 10,000 papers have been published on VRP and its variants, making it one of the most studied problems in operations research.

1.2.1 Problem Statement — CVRP I

The **Capacitated Vehicle Routing Problem (CVRP)** is the canonical and most thoroughly studied member of the VRP family. It serves as the baseline from which all other variants are derived.

Informal Statement of the CVRP

A fleet of identical vehicles of capacity Q is stationed at a *depot*. Each of n geographically dispersed customers has a non-negative demand d_i . The task is to design a set of routes — each starting and ending at the depot — such that:

- Every customer is visited **exactly once** by **exactly one** vehicle.
- The total demand served on any single route does not exceed the vehicle capacity Q .
- The total travel cost (distance or time) is minimised.

The assumption that all vehicles are *identical* and *homogeneous* makes the CVRP the simplest non-trivial VRP. Once this problem is well understood, it becomes the foundation for the richer variants discussed in Section 1.3.

1.2.1 Problem Statement — CVRP II

Graph representation. The CVRP is defined on a graph $G = (V, A)$ where:

- $V = \{0, 1, 2, \dots, n\}$ is the vertex set. Node 0 represents the *depot*; nodes $1, \dots, n$ represent the n *customers*.
- $A = \{(i, j) : i, j \in V, i \neq j\}$ is the arc set. Each arc (i, j) has an associated *travel cost* $c_{ij} \geq 0$.

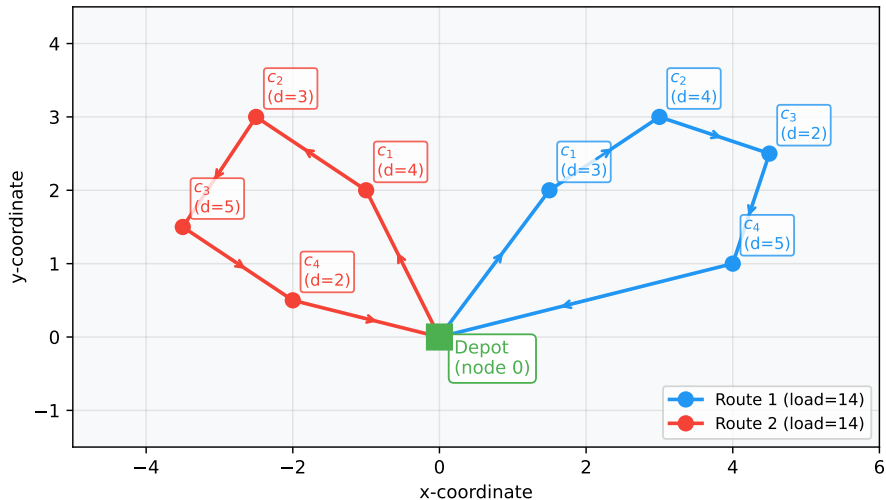
In many practical instances the graph is *symmetric*, i.e. $c_{ij} = c_{ji}$ for all i, j , and the undirected version $G = (V, E)$ is used. The depot plays a special role: every vehicle departs from node 0 at the beginning of its route and returns to node 0 at the end.

Small Instance

Consider $n = 5$ customers with demands $d_1 = 3, d_2 = 4, d_3 = 2, d_4 = 5, d_5 = 3$ and vehicle capacity $Q = 10$. A feasible solution partitions the customers into routes, e.g. Route 1: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$ (load = $9 \leq 10$) and Route 2: $0 \rightarrow 4 \rightarrow 5 \rightarrow 0$ (load = $8 \leq 10$). The total cost is the sum of arc costs along both routes.

1.2.1 Problem Statement — CVRP III

Capacitated VRP (CVRP) — Two-Route Example
Capacity $Q = 15$, each route load $\leq Q$



1.2.2 Mathematical Formulation of the CVRP I

The CVRP can be formulated in several equivalent ways. We present the **two-index vehicle-flow** formulation, which is the most widely used in the literature for exact algorithms.

Sets and parameters.

- $V = \{0, 1, \dots, n\}$: vertex set; node 0 is the depot.
- $V_0 = V \setminus \{0\} = \{1, \dots, n\}$: customer set.
- $A = \{(i, j) : i, j \in V, i \neq j\}$: arc set.
- $c_{ij} \geq 0$: travel cost of arc (i, j) .
- $d_i \geq 0$: demand of customer $i \in V_0$.
- $Q > 0$: vehicle capacity (identical for all vehicles).
- K : number of available vehicles (or a sufficiently large upper bound).

Decision variable. Let $x_{ij} \in \{0, 1\}$ be a binary variable that equals 1 if *some* vehicle travels directly from node i to node j , and 0 otherwise. (In the two-index formulation, the vehicle index is suppressed because all vehicles are identical.)

1.2.2 Mathematical Formulation of the CVRP II

CVRP Two-Index Formulation (VRP₂)

$$(VRP) \quad \min \sum_{(i,j) \in A} c_{ij} x_{ij} \quad (1.1)$$

$$\text{s.t.} \quad \sum_{j \in V, j \neq i} x_{ij} = 1 \quad \forall i \in V_0 \quad (1.2)$$

$$\sum_{i \in V, i \neq j} x_{ij} = 1 \quad \forall j \in V_0 \quad (1.3)$$

$$\sum_{j \in V_0} x_{0j} = K \quad (1.4)$$

$$\sum_{i \in V_0} x_{i0} = K \quad (1.5)$$

$$\sum_{S \subseteq V_0, S \neq \emptyset} \sum_{\substack{i \in S, \\ j \notin S}} x_{ij} \geq \left\lceil \frac{\sum_{i \in S} d_i}{Q} \right\rceil \quad \forall S \subseteq V_0 \quad (1.6)$$

$$x_{ij} \in \{0, 1\} \quad \forall (i, j) \in A \quad (1.7)$$

Compact Formulations and Subtour Elimination I

The exponentially many capacity cuts (1.6) can be replaced by *compact* (polynomial-size) formulations at the cost of tighter LP bounds.

MTZ-type constraints. Introduce a continuous variable $u_i \geq 0$ representing the cumulative load on the vehicle when it *leaves* customer i . Then:

$$u_i - u_j + Q x_{ij} \leq Q - d_j \quad \forall i \in V_0, j \in V_0, i \neq j.$$

When $x_{ij} = 1$ (vehicle travels from i to j), this forces $u_j \leq u_i + d_j - (Q - Q) = u_i + d_j$, i.e. the load increases by d_j after visiting j . The bound $u_i \leq Q$ ensures feasibility. When $x_{ij} = 0$ the constraint is redundant.

This formulation has $O(n^2)$ constraints (polynomial) but gives a *weaker* LP relaxation than the exponential capacity cuts, meaning it requires more branching in a branch-and-bound tree.

Compact Formulations and Subtour Elimination II

Flow-based formulations. An alternative compact approach introduces a *commodity flow* on the network. Let f_{ij} denote the flow (= load) carried on arc (i, j) . Then the capacity and flow-conservation constraints become:

$$\begin{aligned} \sum_{j:(i,j) \in A} f_{ij} - \sum_{j:(j,i) \in A} f_{ji} &= d_i & \forall i \in V_0 \\ d_i x_{ij} &\leq f_{ij} \leq (Q - d_j) x_{ij} & \forall (i, j) \in A. \end{aligned}$$

The first constraint is a flow-conservation law at customer i : the flow entering minus the flow leaving equals d_i (the demand “consumed” at i). The second constraint links the flow variable to the binary routing variable: if the arc is not used ($x_{ij} = 0$), no flow travels on it; if it is used, the flow is between the demand at i and $Q - d_j$.

Key Insight

In practice, the exponential capacity-cut formulation strengthened by dynamically separated cuts outperforms compact formulations for medium and large instances. The compact formulations are preferred for theoretical analysis and for very small instances.

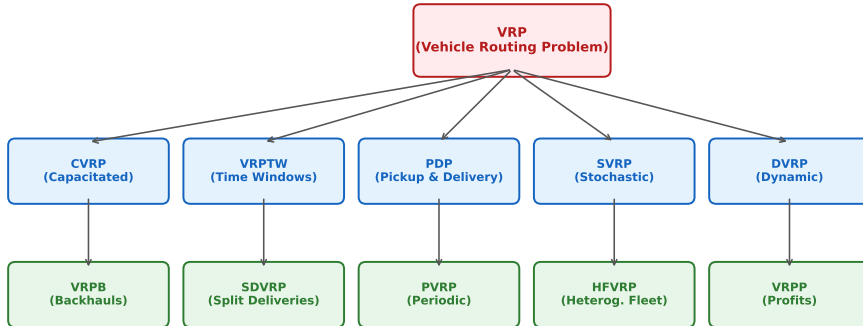
Overview of the VRP Family I

The CVRP is the simplest version of vehicle routing. Real-world applications impose additional constraints or relax some assumptions, giving rise to a rich family of VRP variants. Irnich, Toth, and Vigo organise these along several dimensions:

- ① The **type of transportation request** (delivery, pickup, or pickup-and-delivery).
- ② The **time constraints** (time windows, scheduling, synchronisation).
- ③ The **fleet characteristics** (heterogeneous vehicles, multiple depots, periodic use).
- ④ The **information quality** (deterministic, stochastic, or dynamic data).
- ⑤ The **objective** (cost minimisation, profit maximisation, or bi-criteria).

Overview of the VRP Family II

The Family of Vehicle Routing Problems — Taxonomy Overview



1.3.1 Network Characteristics I

The underlying graph. The network on which the VRP is defined may be *symmetric* (undirected, $c_{ij} = c_{ji}$) or *asymmetric* (directed, $c_{ij} \neq c_{ji}$ possible). Most benchmark instances use symmetric Euclidean distances, but real road networks are asymmetric due to one-way streets, turn restrictions, and direction-dependent traffic.

Distance and Time Matrices. In practice, both a *distance matrix* $D = (d_{ij})$ and a *time matrix* $T = (t_{ij})$ are needed. The distance matrix determines fuel consumption and driving cost, while the time matrix determines whether time window constraints are met. These two matrices need not be proportional to each other (e.g. motorway links are fast but long, while city links are short but slow).

Multiple Depots. In the *Multi-Depot VRP* (MDVRP), vehicles are stationed at several depots. Each vehicle starts and ends its route at the same depot. The optimiser must simultaneously decide which vehicle serves which customers and which depot each vehicle uses.

1.3.2 Types of Transportation Requests I

Delivery and Collection. In the *standard* CVRP the depot delivers goods to customers: the vehicle starts full and empties as it proceeds. In the *Collection VRP* the vehicle starts empty and fills up by collecting items. Both are mathematically equivalent under the capacity constraint, but differ in how route feasibility is checked.

Combined Delivery and Collection. A more complex scenario arises when some customers receive deliveries while others send collections in the same route. There are two sub-variants:

- *VRP with Backhauls (VRPB)*: all deliveries (*linehauls*) must be completed before any collection (*backhaul*) begins, because the vehicle cannot access the rear of a loaded trailer.
- *VRP with Mixed Pickup and Delivery (VRPMPD)*: deliveries and collections can be interleaved freely.

The VRPB constraint that backhauls follow linehauls makes it more restrictive than VRPMPD.

1.3.2 Types of Transportation Requests II

Pickup-and-Delivery (PDP). In the PDP, each transportation request specifies both a *pickup location* p_i and a *delivery location* d_i . The item must be picked up before it is delivered. The same vehicle must handle both ends of a single request (the *coupling constraint*).

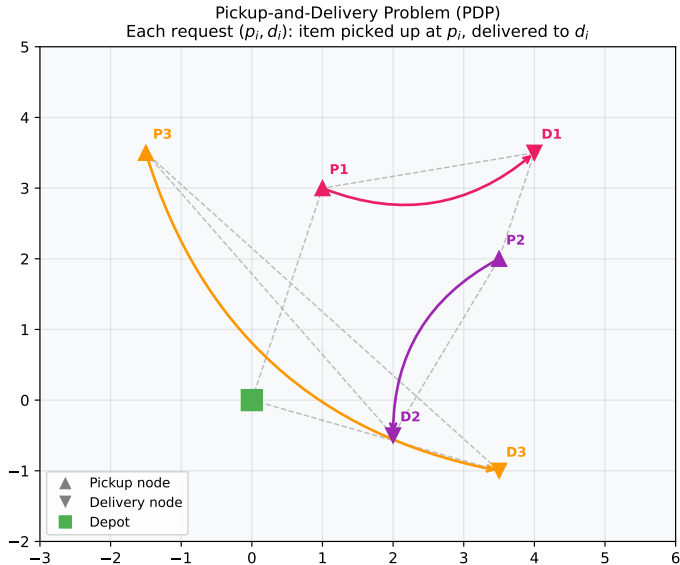
PDP Precedence Constraint

For each request $r = (p_r, d_r)$, if vehicle k serves request r , then p_r must appear *before* d_r in the route of vehicle k :

$$\text{position}(p_r, k) < \text{position}(d_r, k).$$

Important applications of the PDP include passenger transport (*dial-a-ride*), where each passenger has an origin and destination, and container repositioning in shipping logistics.

1.3.2 Types of Transportation Requests III



1.3.3 Time Constraints — VRPTW and Scheduling I

Time Windows. In the *VRP with Time Windows (VRPTW)*, each customer i has a *time window* $[e_i, l_i]$ within which service must begin. Here e_i is the *earliest* acceptable service time (*release time*) and l_i is the *latest* acceptable service time (*deadline*).

VRPTW Time-Window Constraint

Let τ_i denote the time at which the vehicle begins serving customer i . Then:

$$e_i \leq \tau_i \leq l_i \quad \forall i \in V_0.$$

A vehicle arriving *before* e_i may **wait** (waiting is allowed but wasteful). A vehicle arriving *after* l_i renders the solution infeasible.

Time windows arise naturally in food distribution (cold chain requirements), postal delivery (office hours), and medical supply.

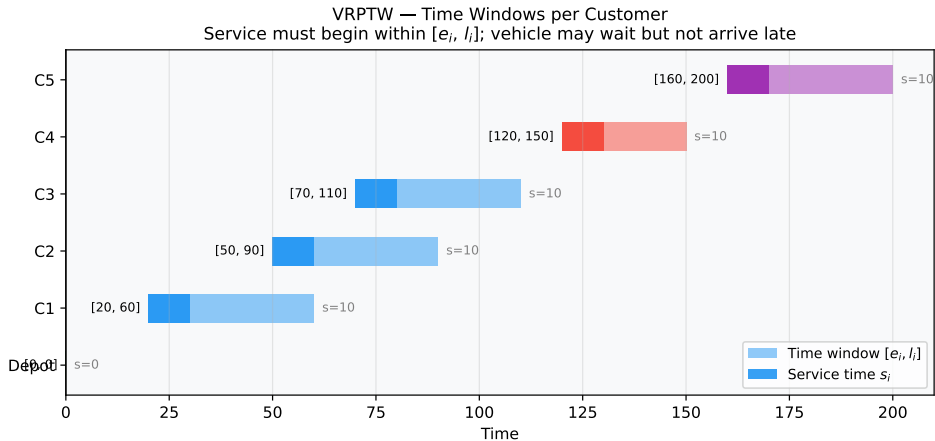
1.3.3 Time Constraints — VRPTW and Scheduling II

Service time. Each customer i also has a service time s_i (time spent unloading or loading at the customer's premises). The departure time from customer i is $\tau_i + s_i$. The arrival time at the next customer j (for a given arc (i, j)) is:

$$\tau_j = \max(e_j, \tau_i + s_i + t_{ij}),$$

where t_{ij} is the travel time on arc (i, j) . The max operator accounts for the possibility that the vehicle arrives before the window opens and must wait.

1.3.3 Time Constraints — VRPTW and Scheduling III



1.3.3 Time Constraints — VRPTW and Scheduling IV

Figure: Time windows for 5 customers. The shaded bars show the feasible service interval $[e_i, l_i]$; the solid portion shows the service time s_i . A vehicle must arrive before the right edge and may arrive (and wait) from the left edge.

1.3.3 Time Constraints — VRPTW and Scheduling V

Hard vs. Soft Time Windows. Time windows are *hard* if violation is strictly forbidden, and *soft* if late arrivals incur a penalty cost rather than being outright infeasible. Soft windows are more realistic in practice (a customer may accept a 15-minute delay with a discount) and allow for more flexible optimisation.

The Depot Time Window. The depot also has a time window $[e_0, l_0]$. The parameter l_0 represents the latest time by which every vehicle must have returned to the depot (e.g. end of the working day or contract requirement). This constraint *limits route duration* and is sometimes called the *route duration constraint*.

Synchronisation. In advanced applications, multiple vehicles must be *synchronised* at the same location or within a time interval. For example, two trucks may need to meet at an intermediate transfer point so that one can hand goods to the other. Synchronisation constraints are expressed as:

$$|\tau_i^k - \tau_i^{k'}| \leq \Delta \quad \text{for vehicles } k, k' \text{ meeting at location } i.$$

TSPTW. When there is only a single vehicle, the VRPTW degenerates to the *Travelling Salesman Problem with Time Windows (TSPTW)*, which is itself NP-hard.

1.3.3 (cont.) Intra-Route Constraints — Loading I

Loading. The *capacity constraint* is the basic intra-route restriction: the total demand served by a vehicle on a route must not exceed Q . More complex loading constraints arise in practice.

Last-In First-Out (LIFO). When the vehicle is a stack (e.g. a container truck with a single opening at the rear), the last item loaded must be the first one unloaded. This is called the *LIFO* constraint. For a delivery route visiting customers c_1, c_2, c_3 in order, the goods for c_3 must be loaded first (deepest in the vehicle), then c_2 , then c_1 (nearest to the door).

3D Loading. In some settings, items have three-dimensional shapes and orientations and must be *physically packed* into the vehicle's cargo space. This gives rise to the *3D-CVRP* or *pallet-packing VRP*, where route design and cargo arrangement are jointly optimised.

Multiple Compartments. *Compartmentalised* vehicles carry incompatible products (e.g. chilled and ambient food) in separate sections. Each compartment has its own capacity, and each customer's order must be allocated to the appropriate compartment.

1.3.3 (cont.) Intra-Route Constraints — Loading II

Key Insight

Loading constraints transform the VRP from a purely graph-combinatorial problem to a problem that also involves a *geometric packing subproblem*. Exact solutions become far harder; heuristics and simulation are typically required.

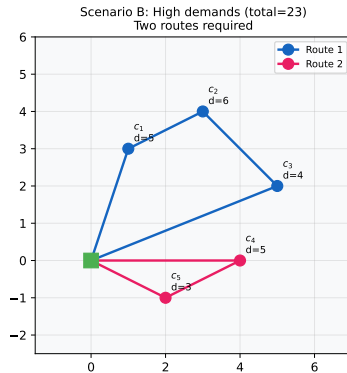
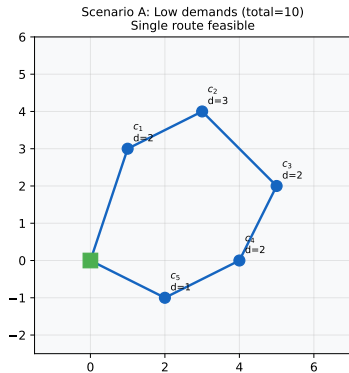
1.3.4 Stochastic VRP (SVRP) I

In the *Stochastic VRP (SVRP)*, some problem data are **not known in advance** but are described by probability distributions. Three main sources of stochasticity occur in practice:

- ① **Stochastic demands:** the demand d_i of customer i is a random variable $\tilde{d}_i$ with known distribution F_i . A route designed for the expected demands may become infeasible if a customer's actual demand exceeds the residual vehicle capacity.
- ② **Stochastic travel times:** traffic congestion makes t_{ij} random; time-window constraints may be violated.
- ③ **Stochastic customer presence:** customer i is present with probability p_i (relevant in “milk-run” delivery or home health-care where not every customer is home).

1.3.4 Stochastic VRP (SVRP) II

Stochastic VRP (SVRP) — Two Demand Scenarios Optimal plan must be robust or allow recourse actions



1.3.4 Stochastic VRP (SVRP) III

Figure: SVRP: two realisations of demand. Under Scenario A (low demands) a single route suffices; under Scenario B (high demands) a second vehicle must be dispatched. The plan must be robust or include a *recourse action*.

1.3.4 Stochastic VRP (SVRP) IV

Modelling approaches for SVRP.

Chance-constrained formulation. A route is accepted if it is capacity-feasible with probability at least $1 - \epsilon$:

$$P\left(\sum_{i \in \text{route}} \tilde{d}_i \leq Q\right) \geq 1 - \epsilon.$$

This is conservative: routes are designed to handle typical (not worst-case) realisations while bounding the probability of a capacity failure.

Stochastic programme with recourse. The plan is a *first-stage* decision; if a capacity violation occurs at runtime, a *recourse action* (e.g. the vehicle returns to the depot for a reload) is taken at an additional cost. The objective minimises expected total cost:

$$\min c(\text{routes}) + \mathbb{E}[\text{recourse cost}].$$

1.3.4 Stochastic VRP (SVRP) V

Key Insight

The SVRP is significantly harder than the deterministic CVRP because the solution quality must be evaluated over a **distribution of scenarios** rather than a single data instance. Scenario-based approaches and robust optimisation are the dominant solution methods.

1.3.5 Dynamic VRP (DVRP) I

The *Dynamic VRP (DVRP)* extends the stochastic setting by allowing the routing plan to be **updated in real time** as new information arrives. While in the SVRP the distribution is known in advance, in the DVRP new orders, cancellations, or traffic events arrive during the execution of routes.

Two types of dynamism:

- *Offline + re-planning*: an initial plan is computed before departure; when disruptions occur, the plan is updated between stops.
- *Fully online*: no advance information; each routing decision is made at the time a vehicle finishes one stop and departs for the next.

Degree of dynamism (*dod*). This metric quantifies what fraction of customer requests arrive dynamically (after the initial plan is committed) versus statically (known before departure). A *dod* of 0 means fully static (deterministic VRP); a *dod* of 1 means fully online (all customers arrive in real time). Real-world courier services have a *dod* around 0.3–0.5.

1.3.5 Dynamic VRP (DVRP) II

Key Insight

For DVRP, solution algorithms must operate under **strict time constraints**: the re-optimisation of routes must complete within seconds so that vehicles receive updated instructions before they reach the next decision point. Fast heuristics and incremental re-optimisation are essential.

1.3.6 VRP with Profits I

In the *VRP with Profits (VRPP)*, the fleet cannot serve all customers within the budget (time, distance, or number of vehicles). Each customer i has a *prize* or *profit* p_i that is collected when the customer is visited. The optimiser must decide **which customers to visit** as well as **how to route the vehicles**, trading off travel costs against collected profits.

Two canonical variants:

- 1 **Team Orienteering Problem (TOP)**: a fixed number of routes, each with a maximum duration; maximise total collected profit.
- 2 **Profitable Tour Problem (PTP)**: routes of unlimited length; maximise *profit minus travel cost*.

VRPP Objective

$$\max \sum_{i \in V_0} p_i y_i - \sum_{(i,j) \in A} c_{ij} x_{ij},$$

where $y_i \in \{0, 1\}$ equals 1 if customer i is visited. Constraints ensure capacity, route feasibility, and that $y_i \leq \sum_j x_{ij}$ (a customer can only earn profit if it is actually visited).

1.3.6 VRP with Profits II

Applications include tourist trip planning, sales-force routing (not all prospects can be visited in a day), and unmanned aerial vehicle (UAV) surveillance routing.

1.3.7 Heterogeneous Fleet VRP (HFVRP) I

In the standard CVRP all vehicles are identical. In practice, a company's fleet almost always consists of vehicles of different types: articulated trucks, rigid trucks, vans, and motorbikes, each with its own capacity, fixed cost, and variable (per-km) cost.

HFVRP Variant Definitions

Let $\mathcal{T}$ be the set of vehicle types. Each type $t \in \mathcal{T}$ has:

- Capacity Q_t (varies by type).
- Fixed cost f_t (paid whenever a vehicle of type t is used).
- Variable cost c_t per unit of distance.
- Number of available vehicles m_t (may be unlimited or bounded).

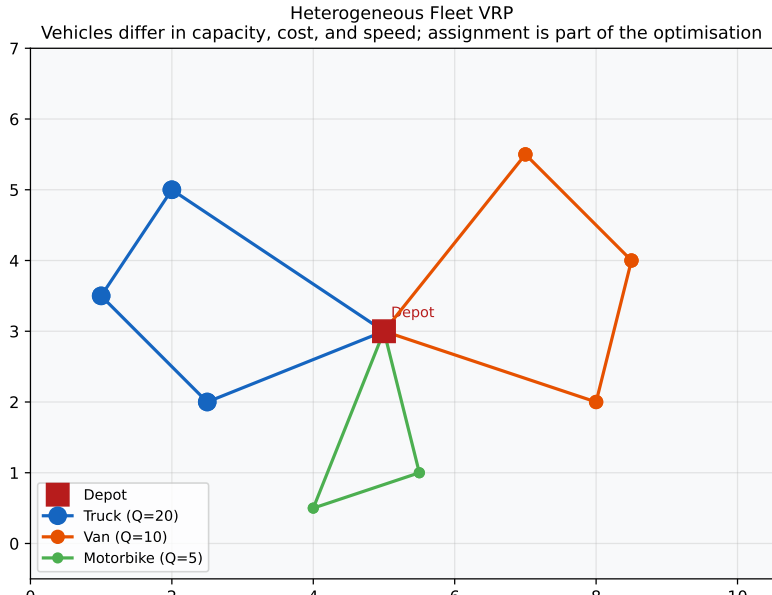
Two sub-variants appear in the literature:

- 1 **Fleet Size and Mix VRP (FSM)**: the number of vehicles of each type is a decision variable (unlimited availability); the goal is to find the optimal fleet composition.

1.3.7 Heterogeneous Fleet VRP (HFVRP) II

- ② **Heterogeneous Fixed Fleet VRP (HFFVRP)**: the fleet composition is fixed (m_t vehicles of each type); only routing is optimised.

1.3.7 Heterogeneous Fleet VRP (HFVRP) III



1.3.8 Periodic VRP (PVRP) I

In the *Periodic VRP (PVRP)*, routing is planned over a **multi-day horizon** $\mathcal{D} = \{1, 2, \dots, T\}$. Each customer i must be visited a certain number of times (frequency) within the horizon, subject to constraints on *which combination of days* is acceptable.

PVRP Service Combination

For customer i , let $\mathcal{S}_i$ be the set of *allowable service patterns* (subsets of days). For example, if $T = 5$ (Mon–Fri), a customer requiring twice-weekly service might accept $\{\text{Mon}, \text{Thu}\}$ or $\{\text{Tue}, \text{Fri}\}$ but not $\{\text{Mon}, \text{Tue}\}$ (too close together). The optimiser must select one pattern from $\mathcal{S}_i$ for every customer and then solve a separate daily routing problem for each day.

Decision levels. The PVRP involves two coupled decisions: (1) assigning service days to customers, and (2) routing vehicles on each day given the day's customer set. These are typically solved with alternating or hierarchical heuristics.

1.3.8 Periodic VRP (PVRP) II

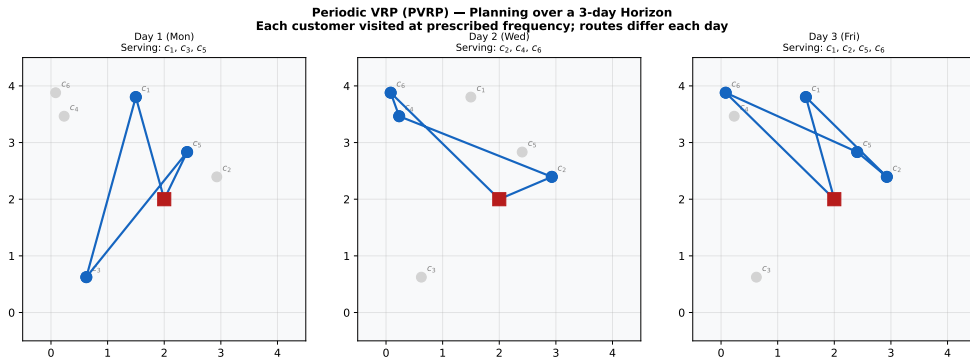


Figure: PVRP over a 3-day horizon. Active customers (blue) differ by day; the routing is re-optimised for each day's customer set.

1.3.3 (cont.) Multi-Use of Vehicles and Routing of Trucks I

Multi-Trip Routing. When the total demand of all customers exceeds what can be delivered in a single loading ($K \cdot Q$), vehicles must make *multiple trips* per day. In the *Multi-Trip VRP (MTVRP)*, a vehicle returns to the depot after completing a trip, reloads, and departs on another trip. The constraint is that the sum of all trip durations must not exceed the maximum working time $T_{\max}$.

Routing of Trucks and Trailers. In the *Truck and Trailer Routing Problem (TTRP)*, not all customer locations are accessible by a full truck-trailer combination (e.g. narrow alleys). At some locations, the truck must decouple the trailer, park it at a designated *satellite* location, and serve the inaccessible customers using the truck alone before returning to re-couple the trailer. This creates a two-level route structure.

Key Insight

Multi-trip and truck-trailer constraints introduce **hierarchical and temporal interdependencies** within a single route that go beyond simple capacity limits. Solving these problems typically requires specialised data structures to track vehicle and trailer states throughout the day.

1.3.9 Other VRP Extensions I

The VRP literature has produced dozens of specialised variants. Here we briefly survey the most important ones not covered in the preceding sections.

Alternative and Indirect Services. Some customers cannot be reached directly from the depot by the delivery vehicle (e.g. remote islands, pedestrian zones). An intermediate facility (*satellite* or *transshipment point*) must be used. This is the *Two-Echelon VRP (2E-VRP)*: large vehicles distribute goods from the depot to satellites (first echelon), and smaller *last-mile* vehicles distribute from satellites to customers (second echelon).

Non-road Transport. The DARP (Dial-a-Ride Problem) models passenger transport where each passenger has a boarding and alighting stop. This is a PDP with the additional constraint that the travel time experienced by each passenger (*ride time*) must not exceed a personal maximum.

Fleet Mix with Green Objectives. *Green VRP* variants minimise CO_2 emissions or fuel consumption rather than distance. Fuel consumption depends on load, speed, and road gradient, so the cost function c_{ij} is no longer fixed but depends on the arc, the vehicle type, and the current load.

Combined Ship Routing. Maritime logistics give rise to the *Ship Routing Problem*, where the “vehicles” are vessels with large capacities and the “road network” is a set of ports connected by sea lanes. Weather windows and port-entry time restrictions add complexity.

1.4 Relationship to the TSP and Bin Packing I

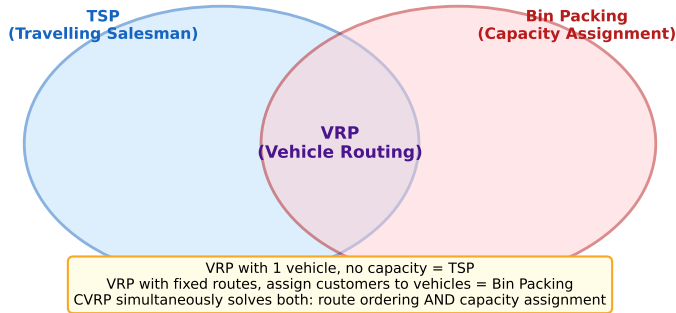
The CVRP combines the combinatorial difficulty of two classical NP-hard problems: the *Travelling Salesman Problem (TSP)* and *Bin Packing (BP)*. Understanding this relationship explains why the CVRP is so hard and suggests algorithmic strategies.

Reduction to TSP. If the vehicle capacity Q is set to ∞ (or at least $\sum_{i=1}^n d_i$), then the capacity constraint is never binding and a single vehicle can serve all customers. The CVRP then reduces to finding the *shortest Hamiltonian cycle* through the n customers and the depot, which is exactly the TSP. Thus $\text{TSP} \subseteq \text{CVRP}$, and any CVRP solver is implicitly a TSP solver.

Reduction to Bin Packing. If all travel costs are set to 0 (or equal), then route ordering does not matter and the only question is: *how to assign customers to vehicles so that each vehicle's total demand does not exceed Q ?* This is equivalent to BP with item sizes d_i and bin capacity Q . The minimum number of routes (vehicles) equals the number of bins in the optimal bin packing. The classical lower bound $K^* \geq \lceil \sum_i d_i / Q \rceil$ comes directly from this analogy.

1.4 Relationship to the TSP and Bin Packing II

VRP as a Generalisation of TSP and Bin Packing



1.4 Relationship to the TSP and Bin Packing III

Figure: VRP as the intersection of TSP and Bin Packing. Special cases of the CVRP reduce exactly to either TSP (no capacity) or Bin Packing (no routing cost).

1.4 Relationship to the TSP and Bin Packing IV

Computational complexity. The CVRP is NP-hard even in the Euclidean case with uniform demands. This follows immediately from the NP-hardness of TSP (which is a special case). In fact, the CVRP is *strongly* NP-hard: no pseudo-polynomial algorithm can solve it unless $P = NP$.

Approximation results. For the metric CVRP (satisfying the triangle inequality), there is a classical $(1 + 2/k)$ -approximation algorithm, where k is the number of customers per vehicle. For the TSP sub-problem, the Christofides–Serdyuk algorithm achieves a $3/2$ approximation. However, for practical problem sizes, exact and heuristic methods substantially outperform these worst-case guarantees.

Exact algorithms. The best exact CVRP solvers (branch-and-cut, branch-and-price, and branch-cut-and-price) can today solve instances with up to ≈ 200 customers to optimality. The LP relaxation of the set-partitioning formulation, strengthened by capacity cuts, provides lower bounds within 1–2% of optimality on standard benchmark instances.

1.4 Relationship to the TSP and Bin Packing V

Key Insight

The dual nature of the CVRP as a routing *and* packing problem is the source of its difficulty and also the source of the best solution methods: the most successful algorithms decompose the problem into a *pricing* sub-problem (route generation via constrained shortest paths) and a *packing* master problem (set partitioning).

1.5 Benchmark Instances for the CVRP and VRPTW I

Standardised benchmark instances are essential for comparing solution methods objectively. Chapter 1 mentions the two most widely used benchmark sets in the VRP literature.

Christofides Instances. The 14 benchmark instances introduced by Christofides, Mingozzi, and Toth (1979) are the oldest and most cited CVRP benchmarks. They range from 50 to 199 customers and are referred to as “CMT1” through “CMT14”. Customer coordinates and demands are taken from distributions that model real delivery scenarios. Several of these instances remained open (no proven optimal solution) until the 2000s.

Solomon Instances. The 56 benchmark instances introduced by Solomon (1987) for the VRPTW are organised into six classes based on two criteria: the spatial distribution of customers (random “R”, clustered “C”, or mixed “RC”) and the tightness of time windows.

1.5 Benchmark Instances for the CVRP and VRPTW II

Solomon Instance Classes

Class	Spatial Distribution	Time Windows
C1, C2	Clustered	Narrow (C1) / Wide (C2)
R1, R2	Random	Narrow (R1) / Wide (R2)
RC1, RC2	Mixed	Narrow (RC1) / Wide (RC2)

Each instance has 100 customers. Narrow-window instances (R1, C1, RC1) tend to require more vehicles with shorter routes; wide-window instances (R2, C2, RC2) allow fewer vehicles with longer routes.

1.5 Benchmark Instances for the CVRP and VRPTW III

Why benchmarks matter. Without standardised instances, claims of improved performance are unverifiable. The VRP community has converged on the Solomon and Christofides sets as the “lingua franca” for comparing algorithms. For each instance, the community maintains a *best-known solution* (*BKS*) table, which is updated as algorithms improve.

Larger benchmarks. For large-scale testing, the *Gehring and Homberger* instances extend Solomon to 200, 400, 600, 800, and 1000 customers. The *Li and Lim* dataset provides PDP benchmarks.

Reading a Solomon Instance File

Each row represents one node. The first row is the depot (node 0). The remaining rows are customers. The columns are: ID, x, y, demand, ready_time (e_i), due_time (l_i), service_time (s_i). For example, a row reading 1 45 68 10 912 967 90 means customer 1 is at position (45,68), has demand 10, with service window [912,967] and a service time of 90 time units.

1.6 Objectives and Performance Metrics I

Single Objective: Minimise Total Distance. The most common objective is to minimise the total travel distance (or cost):

$$\min \sum_{(i,j) \in A} c_{ij} x_{ij}.$$

Distance is a proxy for fuel cost and driver wages. This is appropriate when the number of vehicles is fixed or unlimited.

Hierarchical Objectives. When the fleet size is not fixed, a two-level objective is often used:

- 1 Primary: minimise the number of vehicles (routes) K .
- 2 Secondary: minimise total travel distance given the minimum K .

1.6 Objectives and Performance Metrics II

This reflects reality: an additional vehicle requires a new driver contract, which is far more expensive than a marginal increase in mileage.

Maximise Service Level. When not all customers can be served (e.g. limited fleet in VRPP), the objective switches to maximising the number of served customers (or their total demand, or total profit $\sum_i p_i y_i$).

Secondary Objectives. Route balance (equal workloads among drivers), robustness to demand variability, and carbon emissions are increasingly important secondary objectives in sustainability-oriented applications.

Key Insight

Multi-objective VRP is an active research area. The Pareto front between total distance and number of vehicles is non-trivial because adding a vehicle generally allows significant distance savings by splitting long routes. A knee point on this Pareto curve often provides the most practical solution.

The Clarke-Wright Savings Algorithm I

The **Clarke-Wright savings algorithm** (1964) is the most famous construction heuristic for the CVRP. It starts from a trivially feasible but expensive solution and iteratively merges individual routes to save travel distance, as long as the capacity constraint is not violated.

The saving concept. Suppose we currently have two separate routes: one serving only customer i ($0 \rightarrow i \rightarrow 0$) and one serving only customer j ($0 \rightarrow j \rightarrow 0$). The total cost of these two routes is $c_{0i} + c_{i0} + c_{0j} + c_{j0}$. If we merge them into a single route $0 \rightarrow i \rightarrow j \rightarrow 0$, the new cost is $c_{0i} + c_{ij} + c_{j0}$. The *saving* achieved by this merge is:

$$s(i, j) = c_{0i} + c_{0j} - c_{ij} \geq 0.$$

A positive saving means the merged route is cheaper. The algorithm exploits these savings greedily.

The Clarke-Wright Savings Algorithm II

Worked Example (4 customers, $Q = 10$)

Setup: 4 customers at positions $c_1 = (2, 3)$, $c_2 = (4, 2.5)$, $c_3 = (3.5, 0.5)$, $c_4 = (1, -1.5)$, depot at $(0, 0)$, demands $d_1 = 3$, $d_2 = 4$, $d_3 = 3$, $d_4 = 2$, $Q = 10$.

Step 1 — Initial solution: Four separate routes, each visiting one customer. Total cost $= 2(|0c_1| + |0c_2| + |0c_3| + |0c_4|)$, where $|0c_i|$ is the Euclidean distance from the depot to c_i .

Step 2 — Compute savings: $s(1, 2) = |0c_1| + |0c_2| - |c_1c_2| = 3.61 + 4.72 - 2.06 = 6.27$
 $s(3, 4) = |0c_3| + |0c_4| - |c_3c_4| = 3.55 + 1.80 - 2.24 = 3.11$
 $s(1, 3) = 3.61 + 3.55 - 1.58 = 5.58$, etc.

Step 3 — Sort by saving (descending): $s(1, 2) = 6.27 > s(1, 3) = 5.58 > \dots$

Step 4 — Merge greedily: Merge c_1 and c_2 into route $0 \rightarrow c_1 \rightarrow c_2 \rightarrow 0$ (load $= 7 \leq 10$).

Next, try to merge c_1 and c_3 but c_1 is already interior so only the *endpoint* rule applies. Merge c_3 and c_4 into route $0 \rightarrow c_3 \rightarrow c_4 \rightarrow 0$ (load $= 5 \leq 10$).

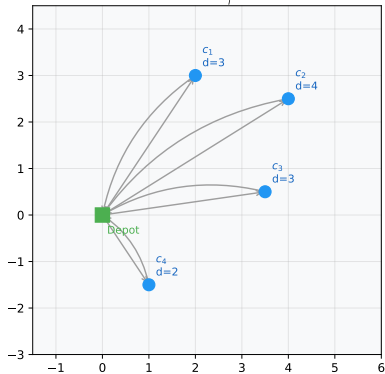
Result: 2 routes, saving achieved $= s(1, 2) + s(3, 4) = 9.38$.

The Clarke-Wright Savings Algorithm III

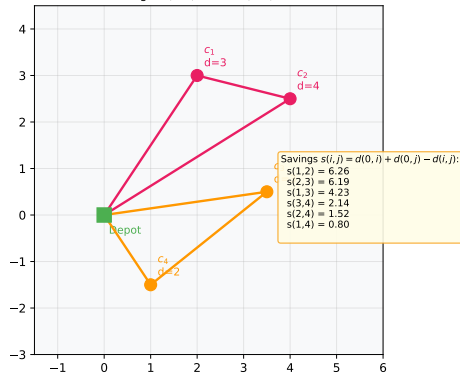
Clarke-Wright Savings Algorithm — Worked Example (Q=10)

Initial: 4 separate routes

$$\text{Total cost} = 2 \sum_i d(0, c_i)$$



After Clarke-Wright: 2 routes
Route 1: $c_1 \rightarrow c_2$ (load=7), Route 2: $c_3 \rightarrow c_4$ (load=5)
Savings: $s(1,2)=6.26$, $s(3,4)=2.14$



The Clarke-Wright Savings Algorithm IV

Figure: Left: initial solution with 4 separate routes. Right: after Clarke-Wright savings, merged into 2 routes. The savings table (bottom right) gives $s(i,j)$ for all pairs.

The Clarke-Wright Savings Algorithm V

Clarke-Wright Savings Algorithm (Pseudocode)

- ① **Initialise:** create n routes, one per customer: $R_i = (0, i, 0)$ for $i = 1, \dots, n$.
- ② **Compute savings:** for all pairs $i < j$, compute $s(i, j) = c_{0i} + c_{0j} - c_{ij}$.
- ③ **Sort:** arrange all pairs in decreasing order of $s(i, j)$.
- ④ **Merge:** for each pair (i, j) in sorted order:
 - If i and j are currently on *different* routes, and i is the *last* customer on its route and j is the *first* customer on its route (or vice versa), and the combined load $\leq Q$:
 - Then merge the two routes by inserting edge (i, j) and removing the depot visits between them.
- ⑤ **Return:** the current set of routes.

The algorithm runs in $O(n^2 \log n)$ time (dominated by the sort step). It produces good solutions quickly but does not guarantee optimality. The *parallel* variant considers all savings simultaneously (as above); the *sequential* variant builds one route at a time.

2-Opt Local Search for Route Improvement I

After constructing an initial solution (e.g. via Clarke-Wright), local search moves are used to improve it. The **2-opt** move is the most fundamental and widely used local search operator for routing problems.

The 2-opt move. A 2-opt move on a single route selects two arcs $(i, i + 1)$ and $(j, j + 1)$ and reverses the sub-sequence of customers between $i + 1$ and j . This is equivalent to removing the two selected arcs and reconnecting the four endpoints in the alternative way that eliminates any crossing.

2-opt Improvement Condition

A 2-opt move on arcs $(i, i + 1)$ and $(j, j + 1)$ is improving if:

$$c_{i,i+1} + c_{j,j+1} > c_{i,j} + c_{i+1,j+1}.$$

That is, the new two arcs are strictly cheaper than the two arcs they replace. If this inequality holds, the move is accepted.

2-Opt Local Search for Route Improvement II

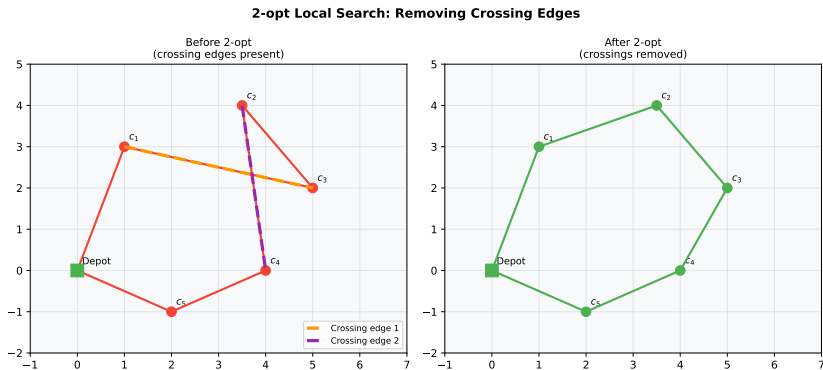


Figure: Left: route with crossing arcs (orange and purple dashed). Right: after 2-opt, the crossing is removed, reducing total distance. The sub-sequence between the two selected nodes is reversed.

2-Opt Local Search for Route Improvement III

2-opt algorithm. The standard 2-opt procedure iterates over all pairs of arcs in a route and applies any improving move, repeating until no improving move exists (a local optimum).

2-opt Algorithm for a Single Route

Input: route $\sigma = (0, \sigma_1, \sigma_2, \dots, \sigma_m, 0)$.

Repeat until no improvement:

- ① Set $\text{improved} \leftarrow \text{False}$.
- ② For each pair $i < j$ with $1 \leq i \leq m-1$, $i+1 \leq j \leq m$:
 - If $c_{\sigma_i \sigma_{i+1}} + c_{\sigma_j \sigma_{j+1}} > c_{\sigma_i \sigma_j} + c_{\sigma_{i+1} \sigma_{j+1}}$: reverse sub-sequence $(\sigma_{i+1}, \dots, \sigma_j)$; set $\text{improved} \leftarrow \text{True}$.
- ③ If $\text{improved} = \text{False}$, stop and return σ .

Complexity: each iteration scans $O(m^2)$ pairs; the number of iterations is at most $O(m^2)$ (each improving move strictly reduces cost), giving overall $O(m^4)$ in the worst case. In practice, far fewer iterations are needed.

2-Opt Local Search for Route Improvement IV

Or-opt. A related move (*Or-opt*) relocates a single customer, a pair, or a triple of consecutive customers to a different position in the same or a different route. Or-opt moves are cheaper to evaluate than 2-opt because they do not require reversing a sub-sequence.

Python: Clarke-Wright and 2-Opt Implementation

```
1 import math, itertools
2
3 def euclidean(a, b):
4     return math.hypot(a[0]-b[0], a[1]-b[1])
5
6 def clarke_wright(coords, demands, Q):
7     """
8     coords : list of (x,y) for depot (index 0) + customers
9     demands : list of demands; demands[0]=0 (depot)
10    Q       : vehicle capacity
11    Returns : list of routes (each route is a list of customer indices)
12    """
13    n = len(coords) - 1          # number of customers
14    depot = coords[0]
15
16    # Compute all savings
17    savings = []
18    for i in range(1, n+1):
19        for j in range(i+1, n+1):
20            s = (euclidean(depot, coords[i]) +
21                 euclidean(depot, coords[j]) -
22                 euclidean(coords[i], coords[j]))
23            savings.append((s, i, j))
24    savings.sort(reverse=True)    # sort descending
25
26    # Initialise: one route per customer
27    route_of = {i: [i] for i in range(1, n+1)}
```

Chapter 1 — Key Takeaways I

This chapter has introduced the foundational vocabulary, models, and taxonomy of the Vehicle Routing Problem family. The following points capture the essential insights.

- 1 **VRP is a universal logistics model.** The generic definition — route a fleet to serve transportation requests at minimum cost — encompasses hundreds of real-world applications from parcel delivery to home health care to maritime logistics.
- 2 **The CVRP is the canonical baseline.** Its two-index formulation with capacity cuts is the starting point for all extensions. The capacity cuts are exponentially many but can be separated efficiently, making branch-and-cut tractable for instances up to ~ 200 customers.
- 3 **The VRP family is vast.** Time windows (VRPTW), pickup-and-delivery (PDP), stochastic demands (SVRP), dynamic requests (DVRP), heterogeneous fleet (HFVRP), periodic planning (PVRP), profit maximisation (VRPP), and split deliveries (SDVRP) are the major variants. Most real instances combine several of these features simultaneously.
- 4 **VRP unifies TSP and Bin Packing.** Removing capacity gives TSP; removing routing costs gives Bin Packing. This dual heritage explains both the hardness (NP-hard) and the algorithmic strategies (column generation decomposes routing from packing).

Chapter 1 — Key Takeaways II

- 5 **Standard benchmarks enable fair comparison.** The Solomon (1987) and Christofides–Mingozi–Toth (1979) instances are the de-facto standards; best-known solutions are tracked by the community.
- 6 **Construction + improvement is the dominant approach.** Clarke-Wright savings constructs a feasible solution in $O(n^2 \log n)$; 2-opt and Or-opt improve it. State-of-the-art solvers use metaheuristics (LNS, ALNS, ILS) built on these primitives.

Final Remark

The rest of the book dives deeply into each variant and each class of solution method. This chapter's models and taxonomy are the common language in which all subsequent chapters are written.

Selected Bibliography I

The following references are cited in Chapter 1 and form the foundational literature of the VRP field.

-  G. B. Dantzig and J. H. Ramser, *The truck dispatching problem*, Management Science, 6 (1959), pp. 80–91. [The original VRP paper.]
-  G. Clarke and J. W. Wright, *Scheduling of vehicles from a central depot to a number of delivery points*, Operations Research, 12 (1964), pp. 568–581. [The savings algorithm.]
-  N. Christofides, A. Mingozzi, and P. Toth, *The vehicle routing problem*, in Combinatorial Optimization, N. Christofides et al., eds., Wiley, Chichester, 1979. [Classical CVRP benchmarks.]
-  M. Solomon, *Algorithms for the vehicle routing and scheduling problem with time window constraints*, Operations Research, 35 (1987), pp. 254–265. [The Solomon VRPTW benchmark instances.]
-  P. Toth and D. Vigo (eds.), *The Vehicle Routing Problem*, SIAM Monographs on Discrete Mathematics and Applications, Philadelphia, 2002. [First edition; predecessor to the present volume.]
-  B. L. Golden, S. Raghavan, and E. A. Wasil (eds.), *The Vehicle Routing Problem: Latest Advances and New Challenges*, Springer, New York, 2008.

Selected Bibliography II



A. Corberán and G. Laporte (eds.), *Arc Routing: Problems, Methods, and Applications*, SIAM, Philadelphia, 2014. [Companion volume on ARP.]



M. Gendreau, G. Laporte, and J.-Y. Potvin, *Logistics*, in *Handbook of Operations Research and Management Science: Transportation*, C. Barnhart and G. Laporte, eds., North-Holland, Amsterdam, 2010.



W. Schmied and R. F. Hartl, *A survey on pickup and delivery models*, *Transportation Science*, 41 (2007), pp. 17–40.

End of Chapter 1

The Family of Vehicle Routing Problems

Irnich · Toth · Vigo

Vehicle Routing: Problems, Methods, and Applications

2nd Edition, SIAM 2014